

Patrick Torbett

STAFF SOFTWARE DEVELOPMENT ENGINEER · TECHNICAL LEAD · MENTOR

San Diego, CA

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Professional Summary

Senior technical leader with deep expertise in large-scale (10k+ transactions per second, multiple collaborating services) secure-by-design abuse prevention systems that balance customer trust with usability in an e-commerce context.

Driving technical strategy and mentoring team leads across 6 teams (60+ engineers), collaborating with senior product/program managers and research/applied machine learning scientists.

Experience

Senior Software Development Engineer (L6)

San Diego, CA

AMAZON, BUYER RISK PREVENTION

July 2021 - Present

- lead team specializing in feature engineering and data access for abuse use cases in collaboration with ML scientists across 50+ evaluation integrations. Integrating and designing ML features for key e-commerce areas, such as returns, refunds, checkout, order fulfillment, customer service, shipment tracking/delivery, payments
- created goals, planned roadmaps, and evolved architecture on signal accuracy, system uptime, and scaling for sales event driven traffic exceeding 10k+ transactions per second
- architected integrations for multiple business models (Zappos, Haul, Global Stores, etc.), developing flexible frameworks that adapt core risk controls to different customer journeys while maintaining security standards.
- collaborated with other tech leads (L6s, L7s) in areas of customer service to design and implement integrations that scale abusive customer contact reductions
- lead technical consultations with product stakeholders for projects to defend against malicious customer service contacts, unify risk signals across the customer journey,
- improved scientist productivity by creating ML feature engineering reference guides for scientists
- scaled mentoring with a focus on team leads and documentation/tooling creation to multiply impact across the organization
- achieved 30% fleet cost reduction and mentored team members in techniques of JVM profiling (CodeGuru Profiler), EC2 instance optimization, and dynamic auto-scaling policies for cost-performance optimization

Software Development Engineer II (L5)

San Diego, CA

AMAZON, BUYER RISK PREVENTION

August 2019 - July 2021

- refurbished and modernized the backend services that support 100s of investigator users that review account data for abuse detection and enforcement (clean code, SOLID principles, Spring MVC)
- replaced monthly release cadence with full CI/CD releases, automated Selenium UI testing, and alerting

Software Development Engineer II (L5)

Seattle, WA

AMAZON, RECommerce

April 2016 - May 2019

- Led the development of "Prophecy", which optimizes selected warehouse process behind 1+ billion decisions for returned/damaged products annually, improving economic recovery of no longer new inventory through algorithmic routing instead of hard-coded business rules
- designed inventory partitioning system to allow the business to palletize liquidation inventory for direct B2B sales instead of truckload level

Software Development Engineer (L4)

Seattle, WA

AMAZON, RECommerce

July 2014 - March 2016

- refactored and maintained end to end invoicing automation for freight-based inventory liquidation shipments
- designed and implemented AWS SNS/SQS event bridge for business event abstraction as part of revamped org architecture

Personal Projects

- (2024) Elixir-based Phoenix LiveView project for home maintenance and pantry tracking with ChatGPT-4o model assisted coding
- (2025) exploratory prototype of Rust 3D graphics using Vulkan rendering with physics and simple shaders using Gemini 2.5 Pro

Education

University of California, San Diego

San Diego, CA

B.S. IN COMPUTER SCIENCE (CUM LAUDE)

2010 - 2014

- specialized coursework in Compilers, Programming Languages (also employed as a tutor for 2 years by UCSD), and Security
- Technical Team Lead on Triton Engineering Student Council
- co-authored conference paper: Chroma: a wearable augmented-reality solution for color blindness (Ubicomp 2014)